Format	Function	Returns
A	Sequence of bytes with trailing nulls and ASCII spaces removed.	String
a	Sequence of bytes.	String
B	Extracts bits from each byte (MSB first).	String
b	Extracts bits from each byte (LSB first).	String
C	Extracts a byte as an unsigned integer.	Fixnum
c	Extracts a byte as an integer.	Fixnum
d,D	Treat sizeof(double) bytes as a native double.	Float
E	Treats sizeof(double) bytes as a double in little-endian byte order.	Float
e	Treats sizeof(float) bytes as a float in little-endian byte order.	Float
f,F	Treats sizeof(float) bytes as a native float.	Float
G	Treats sizeof(double) bytes as a double in network byte order.	Float
g	Treats sizeof(float) bytes as a float in network byte order.	Float
H	Extracts hex nibbles from each byte (most significant first).	String
h	Extracts hex nibbles from each byte (least significant first).	String
I	Treats sizeof(int)° successive bytes as an unsigned native integer.	Integer
i	Treats sizeof(int)° successive bytes as a signed native integer.	Integer
L	Treats four° successive bytes as an unsigned native long integer.	Integer
l	Treats four° successive characters as a signed native long integer.	Integer
M	Extracts a quoted-printable string.	String
m	Extracts a Base64-encoded string. By default, accepts \n and \r. "m0" rejects these.	String
N	Treats four bytes as an unsigned long in network byte order.	Fixnum
n	Treats two bytes as an unsigned short in network byte order.	Fixnum
P	Treats sizeof(char *) bytes as a pointer and returns *len* bytes from the referenced location.	String
p	Treats sizeof(char *) bytes as a pointer to a null-terminated string.	String
Q	Treats eight bytes as an unsigned quad word (64 bits).	Integer
q	Treats eight bytes as a signed quad word (64 bits).	Integer
S	Treats two° bytes characters as an unsigned short in native byte order.	Fixnum
s	Treats two° successive bytes as a signed short in native byte order.	Fixnum
U	Extracts UTF-8 characters as unsigned integers.	Integer
u	Extracts a UU-encoded string.	String
V	Treats four bytes as an unsigned long in little-endian byte order.	Fixnum
v	Treats two bytes as an unsigned short in little-endian byte order.	Fixnum
w	BER-compressed integer (see Array#pack for more information).	Integer
X	Skips backward one byte.	—
x	Skips forward one byte.	—
Z	String with trailing nulls removed.	String
@	Skips to the byte offset given by the length argument.	—
	° May be modified by appending _ or ! to the directive.	

Table 15—Directives for String#unpack

Enumerable

Ruby's Enumerable module is the basis for the functionality of all container classes in Ruby. The most common container classes in use are Array, Hash, and sometimes Set, but this functionality applies to any other class that defines an each method and includes the Enumerable module.

In this section, we will talking about features common to all Enumerables. Other sections in this chapter will talk about how the core implementations of Array, Hash, and Set add their own features.